# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 2 – Industry Problem Solving Task

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|---|---|---|---|
| **Course Code:** | DSA0305 | **Course Title:** | Natural Language Processing |
| **CO Assessed:** | CO4 – Precision | **Assessment:** | Assessment Tool 2 – Industry Problem Solving Task |
| **Weightage:** | 40% of CIA | | |
| **CO5 Statement:** | Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4) | | |
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| **Section / Batch:** | CSE-AI | **Date:** | |

**Code of Conduct**

*I __P.KALYANI (192472199)__ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: P. KALYANI

**Instructions**

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

| Section / Topic | Focus | Question No. |
|---|---|---|
| Representing Meaning & Meaning Structure of Language | Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications. | Q1 |
| Syntax-Driven Semantic Analysis | Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems. | Q2 |

| | | |
|---|---|---|
| First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis | Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy. | Q3 |

**Annexure A – Industry Problem Solving Task**

# 1. Semantic Representation in Customer Support Chatbots

## 1.1 Analysis of Action–Object Relationships

Semantic representation converts a natural-language customer query into a structured meaning representation that can be understood by a chatbot.

| Customer Query | Semantic Representation | Action | Object | Meaning |
|---|---|---|---|---|
| Activate international roaming for my number. | ACTIVATE(Roaming, Customer) | ACTIVATE | Roaming | Customer wants roaming activated |
| Deactivate caller tune service. | DEACTIVATE(CallerTune, Customer) | DEACTIVATE | Caller Tune | Customer wants caller tune deactivated |
| Check my data balance. | QUERY(DataBalance, Customer) | QUERY | Data Balance | Customer wants to check remaining data |
| Enable 5G service. | ACTIVATE(5GService, Customer) | ACTIVATE | 5G Service | Customer wants 5G activated |

The representations follow the general relationship:

**Action → Service/Object → Customer**

For example:

ACTIVATE(Roaming, Customer)

means that the customer is requesting activation of the roaming service.

Semantic representation is important because the chatbot must convert natural-language requests into structured actions that can be connected to telecom backend services.

### 1.2 Identification of Semantic Interpretation Error

The chatbot logs show:

| Query ID | Actual User Intent | Predicted Intent | Result |
|---|---|---|---|
| Q1 | Activate Roaming | Activate Roaming | Correct |
| Q2 | Deactivate Caller Tune | Activate Caller Tune | **Incorrect** |
| Q3 | Check Data Balance | Query Data Balance | Correct |
| Q4 | Enable 5G Service | Activate 5G Service | Correct |

**Q2 contains the semantic interpretation error.**

The correct representation is:

DEACTIVATE(CallerTune, Customer)

However, the chatbot predicts:

ACTIVATE(CallerTune, Customer)

The object is identified correctly, but the action is incorrectly reversed.

This is a serious error because activation and deactivation are opposite operations. If the predicted intent is sent directly to a backend system, the customer's caller tune could be activated instead of cancelled.

### 1.3 Effect of Meaning Representation on Chatbot Decision Accuracy

The four queries contain:

- Correct predictions = 3
- Incorrect predictions = 1

Therefore,

Although the chatbot achieves 75% accuracy, this level may not be sufficient for a service-automation system because a single semantic error can result in an incorrect telecom operation.

For example:

**Correct:**

ACTIVATE(Roaming, Customer)

→ Identify activation request

→ Activate roaming

→ Provide confirmation

**Incorrect:**

ACTIVATE(CallerTune, Customer)

instead of:

DEACTIVATE(CallerTune, Customer)

→ Wrong backend operation

→ Customer dissatisfaction

→ Additional support request

Thus, semantic representation directly influences the accuracy and reliability of chatbot decision-making.

### 1.4 Recommended Improvements

The telecom company should implement the following improvements:

1. **Intent Classification** – Distinguish activate, deactivate, query, cancel and renew requests.
2. **Context-Aware Semantic Analysis** – Use previous conversation information.
3. **Negation Detection** – Detect terms such as *not, deactivate, cancel, stop,* and *disable*.
4. **Entity Recognition** – Identify services such as roaming, 5G and caller tune.
5. **Domain-Specific Ontology** – Maintain relationships between telecom services and possible actions.
6. **Confidence Scoring** – Assign a confidence value to each predicted intent.
7. **Confirmation Mechanism** – Ask the customer to confirm critical service operations when confidence is low.
8. **Continuous Learning** – Use corrected customer interactions as training data.

For example:

“Do you want me to deactivate your caller tune service?”

can be presented before performing a potentially incorrect operation.

**Conclusion**

Semantic representation provides a bridge between natural language and chatbot actions. The given data shows 75% prediction accuracy, but Q2 demonstrates the risk of confusing opposite

actions. Therefore, intent classification, contextual analysis, negation detection, entity recognition and confirmation mechanisms should be combined to improve telecom chatbot reliability.

## 2. First-Order Predicate Calculus for Smart Manufacturing

### 2.1 Representation of Production Data Using First-Order Predicate Calculus

The production data is:

| Machine | Status |
|---|---|
| M1 | Active |
| M2 | Active |
| M3 | Maintenance |
| M4 | Active |

The facts can be represented as:

Active(M1)

Active(M2)

Maintenance(M3)

Active(M4)

**Predicate Rules**

**Rule 1:**

Therefore:

Producing(M1)

Producing(M2)

Producing(M4)

**Rule 2:**

This means that if an active machine produces a product, that product is considered available.

**Rule 3:**

Therefore:

¬Producing(M3)

### 2.2 Analysis of Currently Available Products

The exact products currently available cannot be determined from the supplied data because there is no machine-to-product relationship.

The missing information would be represented as:

Produces(M1,Gear)

If:

Produces(M1,Gear)

Active(M1)

then:

Available(Gear)

can be inferred.

Therefore, the correct conclusion is:

Specific available products cannot be determined because the Produces(x,y) relationships are not provided in the dataset.

This is an important data limitation because predicate logic can only infer conclusions from available facts and rules.

**2.3 Effect of Maintenance on Gear Production**

The given data establishes:

Maintenance(M3)

Therefore, according to the rule:

we obtain:

¬Producing(M3)

If M3 produces Gear:

Produces(M3,Gear)

then Gear production by M3 is affected by maintenance.

However, the dataset does not explicitly state:

Produces(M3,Gear)

Therefore, it is not logically valid to conclude that overall Gear production is affected.

**Conditional Conclusion**

If M3 is responsible for producing Gear, its maintenance activity stops Gear production on M3.

This demonstrates the importance of complete production data for reliable industrial reasoning.

**2.4 Effectiveness of Predicate Logic in Industrial Decision Support**

First-Order Predicate Calculus provides several advantages.

**Advantages**

1. **Precise representation** – Machine states can be represented formally.
2. **Rule-based inference** – New facts can be derived automatically.
3. **Explainability** – Decisions can be traced back to logical rules.
4. **Maintenance monitoring** – Machines under maintenance can be identified.

5. **Automation** – The same rules can be continuously applied to new production data.

For example:

and:

**Limitations**

Predicate logic alone does not handle:

- uncertain information
- noisy sensor readings
- probabilistic failures
- predictive maintenance
- machine-learning-based forecasting

Therefore, a practical Industry 4.0 system can combine predicate logic with IoT sensors, databases, machine learning and rule engines.

**Conclusion**

Predicate calculus provides an explainable and systematic framework for manufacturing decision support. However, the current dataset lacks machine-to-product relationships. Therefore, additional production mapping is required to determine exact product availability and Gear-production impact.

## 3. Word Sense Disambiguation in E-Commerce Search Engines

### 3.1 Determination of Correct Word Sense

The search context and clicked results provide sufficient evidence to determine the intended meaning.

| User Query | Possible Sense 1 | Possible Sense 2 | Clicked Result | Correct Sense |
|---|---|---|---|---|
| Apple accessories | Fruit | Technology Brand | iPhone Charger | **Technology Brand** |
| Mouse wireless | Animal | Computer Device | Bluetooth Mouse | **Computer Device** |

| | | | | |
|---|---|---|---|---|
| Java tutorial | Island | Programming Language | Coding Lessons | **Programming Language** |
| Python course | Snake | Programming Language | Software Development Training | **Programming Language** |

The clicked results provide strong evidence that the users are searching for technology-related meanings.

### 3.2 Semantic Cues Used for Disambiguation

**Apple accessories**

The word "accessories" suggests a product-related context. The clicked result iPhone Charger confirms the technology-brand meaning.

**Mouse wireless**

The term "wireless" is strongly associated with electronic devices. The clicked Bluetooth Mouse confirms that the intended meaning is a computer device.

**Java tutorial**

The term "tutorial" provides an educational context. The clicked result Coding Lessons confirms that Java refers to the programming language.

**Python course**

The word "course" combined with software development training indicates that Python refers to the programming language.

The major semantic cues are:

- neighboring words
- query intent
- product category
- clicked results
- historical user behavior
- domain context

### 3.3 Impact of Incorrect Sense Selection

Incorrect Word Sense Disambiguation can produce irrelevant search results.

**Example: Apple accessories**

If **Apple** is incorrectly interpreted as a fruit, the system may return:

- fruit baskets
- apple recipes
- agricultural products

instead of:

- iPhone chargers
- AirPods accessories

- Apple Watch accessories

**Example: Python course**

If Python is interpreted as a snake, the system may return:

- reptile information
- snake-care courses
- animal-related products

instead of programming courses.

Therefore:

**Wrong Sense → Irrelevant Results → Lower Relevance → Fewer Clicks → Lower Conversion → Poor User Satisfaction**

For a large e-commerce company, this can directly affect product discovery, sales and customer retention.

### 3.4 Industrial-Scale WSD Strategy

A hybrid contextual WSD system can be designed as:

User Query

↓

Tokenization

↓

Context Analysis

↓

Named Entity Recognition

↓

Candidate Sense Generation

↓

Contextual Embedding Model

↓

User/Search History

↓

Sense Ranking

↓

Product Retrieval

↓

Recommendation Ranking

**Recommended Technologies**

1. **Contextual Embeddings** – Understand word meaning according to surrounding words.
2. **Product Ontology** – Represent different meanings and product categories.
3. **Search History** – Use previous user behavior to identify intent.
4. **Click-Through Data** – Use clicked results as feedback.
5. **Feedback Learning** – Learn from clicks, purchases, search reformulation and product views.

For example, an ontology can represent:

```
Apple
 ├── Technology Brand
 └── Fruit

Mouse
 ├── Computer Device
 └── Animal
```

**Conclusion**

The data demonstrates that context is the key factor in resolving ambiguous words. An industrial WSD system combining contextual models, product ontology and user behavior can improve search relevance, recommendations, customer satisfaction and conversion opportunities.

## 4. Syntax-Driven Semantic Analysis in Healthcare Information Systems

### 4.1 Contribution of Syntactic Structures to Semantic Role Assignment

The healthcare system processes:

1. Doctor prescribed medicine to patient.
2. Patient reported severe headache.
3. Nurse monitored patient continuously.
4. Medicine reduced blood pressure.

Syntactic structure helps determine relationships between entities.

For example:

**Doctor prescribed medicine to patient.**

Structure:

**Subject → Verb → Object → Recipient**

Therefore:

- Doctor → Agent
- Medicine → Theme/Object

- Patient → Recipient

Similarly:

**Patient reported severe headache.**

- Patient → Experiencer/Agent
- Headache → Symptom

Thus, syntax provides the structural foundation for semantic-role labeling.

**4.2 Evaluation of Assigned Semantic Roles**

The given assignments are:

| Entity | Assigned Role | Evaluation |
|---|---|---|
| Doctor | Agent | **Appropriate** |
| Medicine | Instrument | **Not appropriate** |
| Patient | Recipient | **Appropriate for first sentence** |
| Headache | Symptom | **Appropriate** |

**Correction**

In:

**“Doctor prescribed medicine to patient.”**

Medicine should be assigned the role Theme/Object, not Instrument.

An instrument is something used to perform an action.

Example:

“The doctor used a syringe.”

Here:

- Doctor → Agent
- Syringe → Instrument
- Medicine → Theme
- Patient → Recipient

Therefore, Medicine = Instrument is a semantic-role error.

### 4.3 Potential Errors Due to Incorrect Parsing

Incorrect parsing can result in incorrect clinical interpretation.

Correct interpretation:

Doctor → prescribed → Medicine

Doctor → prescribed → Medicine → to Patient

An incorrect parser could potentially interpret:

Patient → prescribed → Medicine → to Doctor

This changes the complete meaning of the medical statement.

**Possible consequences**

- Incorrect medical records
- Wrong treatment extraction
- Incorrect patient history
- Incorrect clinical decision support
- Incorrect alerts
- Reduced doctor trust
- Potential patient-safety risks

Therefore, healthcare NLP requires a higher level of semantic accuracy than ordinary information-retrieval applications.

### 4.4 Recommended Methods for Improvement

**1. Dependency Parsing**

Dependency parsing identifies grammatical relationships between words.

Example:

Doctor
  ↓ nsubj
prescribed
  ↓ dobj
medicine
  ↓ nmod
patient

This provides detailed relationships between medical entities.

**2. Semantic Role Labeling**

The system should identify roles such as:

- Agent

- Patient/Theme
- Recipient
- Experiencer
- Instrument
- Location
- Time

**3. Medical Named Entity Recognition**

The system should recognize:

- Doctor
- Patient
- Medicine
- Disease
- Symptom
- Treatment
- Hospital
- Location

**4. Medical Ontology**

Domain knowledge should distinguish:

Medicine ≠ Instrument

Patient ≠ Treatment

Disease ≠ Medication

5. Domain-Specific Parsing

Generic NLP parsers may not understand specialized medical terminology. Medical corpora and domain-specific models should therefore be used.

**6. Hybrid Syntax-Semantic Architecture**

A practical architecture is:

Medical Text

↓

Preprocessing

↓

POS Tagging

↓

Dependency / CFG Parsing

↓

Feature Validation

↓

Semantic Role Labeling

↓

Medical NER

↓

Ontology Mapping

↓

Relation Extraction

↓

Clinical Knowledge Graph

**Conclusion**

Syntax provides the basic structure, but syntax alone is insufficient for accurate medical interpretation. Combining dependency parsing, semantic role labeling, medical NER, domain ontology and relation extraction can significantly reduce semantic errors and improve the reliability of healthcare NLP systems.

## Rubrics for Calculation

| **Criteria** | **Weight age** | **Level 4 – Exemplary** | **Level 3 – Proficient** | **Level 2 – Developing** | **Level 1 – Beginning** |
|---|---|---|---|---|---|
| Understanding of Industry Problem | 20% | Comprehensive understanding of real-world problem and constraints | Good understanding with minor gaps in requirements | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |
| Application of Engineering Concepts | 25% | Applies advanced and relevant concepts effectively to solve the problem | Applies appropriate concepts with minor errors | Limited application of concepts | Incorrect or no application of concepts |
| Solution Design | 25% | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps | Basic solution with limited feasibility | Unclear or impractical solution |

| | | | | | |
|---|---|---|---|---|---|
| Use of Tools | 15% | Effective use of modern tools, technologies, and real data | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools | Minimal or incorrect use of tools |
| Analysis | 10% | Thorough analysis with validated results and performance evaluation | Good analysis with reasonable validation | Basic analysis with limited validation | Poor or no analysis |
| Professionalism | 5% | Highly professional, well-documented, and clearly presented work | Good documentation and presentation | Basic documentation with some gaps | Poor documentation and presentation |

| Q. No. / Criteria Level | Understanding of Industry Problem | Application of Engineering Concepts | Solution Design | Use of Tools | Analysis | Professionalism | Sub. Total | Total % |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | |
| 2 | | | | | | | | |
| 3 | | | | | | | | |

| Faculty In-charge | Course Coordinator | Program Director |
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| Signature: _______________ | Signature: _______________ | Signature: _______________ |
| Date: ___________________ | Date: ___________________ | Date: ___________________ |